

KESARI DASARADHA RAMI REDDY

Python Backend Engineer | FastAPI · PostgreSQL · Async APIs

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PROFESSIONAL SUMMARY

Backend-focused **Python Backend Engineer** with nearly **3 years of experience** designing and building **scalable, high-performance backend systems** using Python, FastAPI, SQLAlchemy, and PostgreSQL. Strong expertise in **asynchronous REST API development**, modular backend architecture, and secure authentication and authorization using **OAuth2, JWT (RS256), and RBAC**. Proven track record of optimizing database performance, improving system reliability, and delivering production-grade services with measurable impact, including significantly faster deployments and increased release velocity in cloud-native environments.

PROFESSIONAL EXPERIENCE

Python Backend Engineer

KaryaHub Solutions

August 2023 – Present

Hyderabad, India

- Led backend development across multiple production platforms (HRMS, ERP, ticketing), designing and maintaining **100+ asynchronous REST APIs** with consistent sub-200ms latency.
- Re-architected a tightly coupled monolithic backend into **Dockerized modular services** behind **NGINX reverse proxy**, reducing deployment duration from **2 hours to 15 minutes (87% improvement)** and increasing horizontal scalability by 40–50%.
- Engineered and deployed a secure authentication layer leveraging **OAuth2, JWT (RS256), refresh/revoke token strategies, and fine-grained RBAC**, combined with **AES-256 (Fernet) encryption** for sensitive data handling aligned with GDPR and PCI-DSS practices.
- Improved **PostgreSQL schema design, indexing, and complex query performance**, and executed zero-downtime migrations across staging and production environments using **Alembic**.
- Built and maintained **CI/CD pipelines** using GitHub Actions, integrating **SonarQube, Snyk, Trivy, Bandit, and Dependabot** to detect 30% more critical issues before production release and enabling a 40% faster release cadence.
- Integrated **HashiCorp Vault** for centralized secrets management, removing hardcoded credentials and strengthening security posture by approximately 45%.
- Introduced **Redis-based caching and API rate limiting** mechanisms to prevent brute-force attacks and protect public endpoints from abuse.
- Provisioned and managed **GCP infrastructure** including Compute Engine, Cloud SQL (PostgreSQL), VPC networking, IAM roles, Cloud NAT, Managed Instance Groups, and Load Balancers, using **Terraform (Infrastructure as Code)**.
- Owned comprehensive API documentation via **Swagger/OpenAPI and MkDocs**, improving cross-team onboarding efficiency by 35% and minimizing integration bottlenecks.
- Collaborated with frontend engineers, designers, and product stakeholders to define API contracts, validate technical feasibility, and deliver backend solutions aligned with product timelines.
- Mentored 3 junior developers on asynchronous Python design patterns, Git workflows, and API security best practices, reducing onboarding time by 35%.

PROJECTS

• Neuropi — School Management ERP

Python · FastAPI · Async SQLAlchemy · PostgreSQL · Redis · Docker · Pydantic v2 · Alembic

- Developed **100+ asynchronous REST APIs** across admissions, student management, attendance, library, and document workflows serving **5,000+ students**.
- Refactored a legacy synchronous backend into an **async-first architecture**, resolving event-loop issues and improving deployment stability by **50%**.
- Implemented **fine-grained RBAC** with strict request/response validation using Pydantic v2, reducing data inconsistencies by **95%**.
- Introduced Redis-based caching and rate limiting to improve response times and protect authentication endpoints.
- Built environment-aware file storage handling for local and cloud deployments with metadata persisted in PostgreSQL.

• SHOU — HRMS and Animation Production Pipeline Tracker

Python · FastAPI · PostgreSQL · Docker · NGINX · Alembic · SonarQube · Snyk

- Engineered a comprehensive **HRMS backend** handling payroll, attendance, leave, and reporting modules, processing **5,000+ monthly payroll records with 100% accuracy**.
- Designed an animation production pipeline tracker to manage project phases, asset assignments, and milestone workflows for creative teams.
- Migrated a legacy backend into a **Docker-based modular architecture**, significantly improving deployment reliability and scalability.
- Strengthened authentication using JWT lifecycle management and RBAC, reducing unauthorized access attempts by **99%**.
- Integrated automated security scanning and enforced zero-downtime database migrations using Alembic.

• Events2Go — Event Discovery and Ticketing Platform

Python · FastAPI · Async SQLAlchemy · PostgreSQL · Redis · Docker · AWS S3 · Stripe

- Designed and built a high-concurrency **async backend system** supporting **10,000+ concurrent users** with event discovery, venue management, and multi-stage booking workflows.
- Engineered a **concurrency-safe seat reservation engine** using PostgreSQL transactions and row-level locking, completely eliminating double-booking under peak traffic.
- Implemented a **multi-tenant RBAC model** for platform admins, organizers, and end users, enforced through Pydantic v2 validation.
- Built secure **JWT-based authentication flows** with access, refresh, and revoke tokens, including device-level session tracking.
- Integrated **Stripe payments** with encrypted workflows, ensuring PCI-DSS-aligned payment processing.
- Reduced database load by **35%** through async query optimization and Redis caching while maintaining sub-200ms API latency.
- Authored **50+ async Pytest** test cases covering booking logic, payment edge cases, and concurrency scenarios.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Frameworks & Libraries:** FastAPI (sync/async), SQLAlchemy ORM, Pydantic v2
- **Backend Engineering:** RESTful API Design, Async Programming, Modular Architecture, Request/Response Validation, Swagger/OpenAPI, MkDocs
- **Databases & Caching:** PostgreSQL (schema design, indexing, query optimization), Redis (caching, rate limiting), Alembic
- **Authentication & Security:** OAuth2, JWT (HS256/RS256), RBAC, AES-256 Encryption (Fernet), HashiCorp Vault
- **DevOps (Working Knowledge):** Docker, GitHub Actions, NGINX, Caddy, Git
- **Cloud Platforms:** Google Cloud Platform (Compute Engine, Cloud SQL, VPC), DigitalOcean Spaces
- **Testing & Code Quality:** Pytest (unit, integration, async), Postman, Black, isort, Flake8, Pylint, SonarQube, Snyk, Trivy, Bandit

EDUCATION

• Bachelor of Technology (B.Tech) — Electrical & Electronics Engineering

Vasireddy Venkatadri Institute of Technology (VVIT)

2018 – 2022

Guntur, India